

to carry data that doesn't need to be modified. The instructions carried on the ROM encompass the firmware, which provides an identity to the device and work closely with its associated hardware. ROM being non-volatile, firmware is an 'ever-present' data - it won't be lost if the device is switched off or restarted. Hence the "firm". Of course, it does much more than that. Firmware coordinates the various operations happening within the device (which is why, your OS can be called a firmware). There are three types of these when it comes to mobile devices: stock firmware, manufacturer branded firmware and custom firmware/ROM.

Google's original Android can be considered a stock firmware, which sees its own set of updates from the OS manufacturer. Some examples of manufacturer branded firmware for mobile devices include HTC's Sense, Samsung's Touch Wiz, and many more. In these cases, the manufacturer has overlaid their own user interface and design over the default Android architecture, giving it a unique flavour. Custom firmware, though, is fully designed by third party developers. It replaces the stock firmware, and while it may be Android, it provides an entirely new feel and usually, an upgraded set of features.

However, if it can't be changed, what if one day, firmware becomes a limiting factor? For that matter, if you don't keep your firmware up-to-date, there's a chance you could face security lapses, and miss out on improvements to pre-existing features (because like PC software, no firmware is perfect at launch) or simply not take full advantage of the hardware in your hands.

That is where flashing comes in.

What is Flashing?

Flashing basically means sending your phone to Flash, because the Justice League doesn't provide a company phone these days. And may God be with you if you can believe that.

When adding new features to existing data on ROM, manufacturers tend to overwrite certain modules. This is very much like spring cleaning: removing the old data to make way for the new. The entire process is known as "flashing firmware" or simply flashing. This is usually done by connecting your mobile device to a PC via USB, and using the software provided by a manufacturer. You can also employ software like YourCellular, though instructions may vary for specific phones. It's quite simple and the entire process takes about 15-20 minutes.